Unit 16: Short-Run Costs

I. Short-Run Cost Measures

II. Short-Run Cost Curves

III. Production Functions and the Shape of Cost Curves

IV. Effects of Taxes on Costs

Objectives of the Study

After studying this unit, the student will be able to

- Understand the short run cost measures
- Distinguish between different cost curves
- Acquire knowledge about the effects of taxes on costs

I. Short-Run Cost Measures

II. Short-Run Cost Curves

In microeconomic theory, cost analysis plays a central role in understanding firm behaviour. A profit-maximizing firm must carefully examine how its costs change with variations in output. Since profit is defined as the difference between total revenue and total cost, understanding the structure and behaviour of costs is essential for rational decision-making. In the short run, firms face certain constraints. Some inputs, particularly capital such as machinery, buildings, and land, cannot be adjusted immediately. Because of these fixed inputs, increasing output in the short run often becomes more expensive compared to the long run, where all inputs are variable.

The short run is defined as a time period during which at least one factor of production is fixed. Typically, capital is fixed in the short run, while labor and raw materials are variable. For example, a manufacturing firm may not be able to immediately expand its factory size or purchase new heavy machinery. But, it can increase output by hiring more workers or using

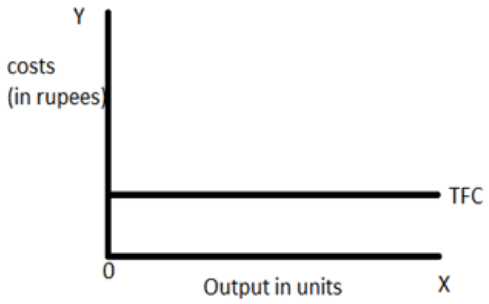
more raw materials. Because some inputs are fixed, the firm cannot freely adjust its scale of operation. This limitation affects cost behaviour. When output increases beyond a certain level, the firm may face rising costs due to the law of diminishing returns, which states that adding more units of a variable factor to fixed factors eventually reduces marginal productivity.

Short-run costs are classified into several categories:

Fixed Cost (F):

Fixed cost refers to those production expenses that do not change with the level of output in the short run. These costs remain constant whether the firm produces nothing at all or operates at full capacity. Because at least one factor of production—usually capital—is fixed in the short run, the expenses associated with that factor are also fixed. A firm must pay these costs regardless of its production decisions. Examples of fixed costs include rent for the factory building, insurance premiums, salaries of permanent administrative staff, depreciation of machinery and equipment, interest on borrowed capital and property taxes. These costs are incurred even if the firm temporarily stops production. Another key characteristic of fixed cost is that, although total fixed cost remains constant, the average fixed cost (AFC) decreases as output increases. This happens because the same fixed cost is spread over more units of production. Thus, higher output reduces fixed cost per unit, even though total fixed cost does not change. Graphically, the fixed cost curve is represented by a horizontal straight line as shown in the figure 16.1.

Figure 16.1: Fixed Cost Curve

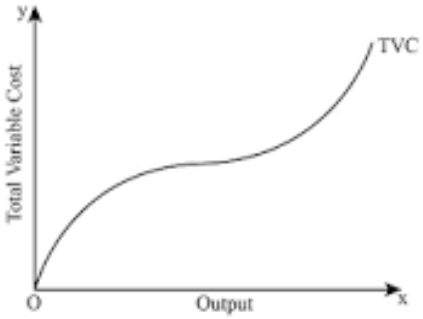


Since fixed cost remains the same at all levels of output, the curve does not slope upward or downward. It runs parallel to the horizontal axis at the level of the fixed cost amount. This graphical representation clearly shows that output has no effect on fixed cost in the short run.

Variable Cost (VC)

Variable cost refers to those production expenses that change directly with the level of output. Unlike fixed costs, variable costs increase when output increases and decrease when output decreases. These costs are associated with variable inputs—inputs that a firm can adjust in the short run to change its level of production. Common examples of variable costs include wages paid to temporary or daily wage workers, cost of raw materials, electricity and fuel used in production, packaging expenses, transportation costs related to output etc. Graphically, the variable cost curve starts at the origin (because when output is zero, variable cost is zero) and slopes upward as shown in figure 16.2.

Figure 16.2: Variable Cost Curve



Initially, it may rise gently, but after a certain point, it becomes steeper due to diminishing returns.

Total Cost (C)

Total cost refers to the overall cost incurred by a firm in producing a given level of output in the short run. It includes both fixed costs and variable costs. In other words, total cost represents the complete economic sacrifice made by the firm to produce goods or services. The relationship is expressed as:

$$C = VC + F$$

Where,

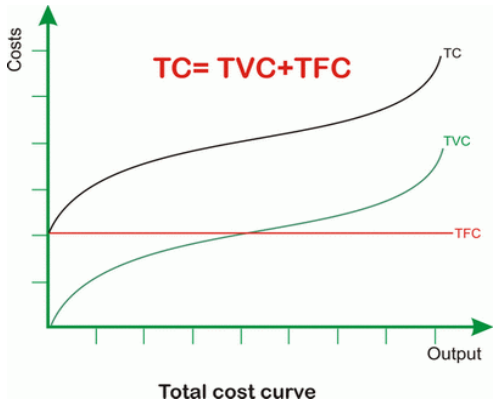
C = Total Cost

VC = Variable Cost

F = Fixed Cost.

Any change in total cost is caused entirely by changes in variable cost because fixed cost remains constant in the short run. As output increases, more labor, raw materials, electricity, and other variable inputs are required. Therefore, variable cost rises, and total cost also increases. In the initial stages of production, total cost may increase at a decreasing rate due to better utilization of fixed factors and improved efficiency. But, after a certain point, diminishing marginal returns set in. At this stage, additional units of output require increasingly larger additions to variable inputs, causing total cost to increase at an increasing rate. The graphical representation of the total cost curve is shown in the figure 16.3.

Figure 16.3: Total Cost



The Fixed Cost (F) curve is a horizontal straight line. The Variable Cost (VC) curve starts from the origin and slopes upward. So, the Total Cost (TC) curve starts from the fixed cost level (not from zero) and slopes upward. The total cost curve is parallel to the variable cost curve because: TC - VC = F. Since fixed cost is constant at all output levels, the vertical distance between the total cost curve and the variable cost curve is always equal to fixed cost. This constant vertical gap explains why the two curves are parallel.

Marginal Cost (MC)

Marginal cost is the additional cost incurred by a firm when it produces one more unit of output. It measures how total cost changes as output changes by a small amount, usually one unit.

Marginal cost is one of the most important cost concepts in microeconomics because it directly guides production decisions. The formula for marginal cost is

$$MC = \Delta C / \Delta q$$

Where,

ΔC = Change in total cost

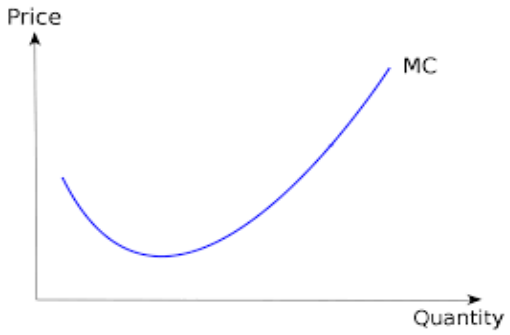
Δq = Change in output.

Since fixed cost does not vary with output in the short run, any change in total cost is entirely due to a change in variable cost. Therefore, marginal cost can also be expressed as:

$$MC = \Delta VC/\Delta q$$

This means marginal cost reflects the additional variable cost required to produce one more unit. Marginal cost plays a central role in profit maximization. The basic profit-maximizing rule is when MR = MC. A firm compares marginal cost (MC) with marginal revenue (MR), which is the additional revenue from selling one more unit for profit maximisation. Shape of the marginal cost curve is shown in the figure 16.4.

Figure 16.4: Marginal Cost Curve



The marginal cost curve is typically U-shaped. This shape is explained by the law of diminishing marginal returns. Marginal cost curve is having two stages. Initially decreasing and later increasing. Initial Stage (Decreasing MC) where when production begins, marginal cost may fall. This happens because fixed factors such as machinery are better utilized. Workers specialize, coordination improves, and productivity increases. As a result, each

additional unit of output costs less to produce. Later Stage (Increasing MC) where after a certain level of output, diminishing returns set in. Since some inputs are fixed, adding more variable inputs leads to overcrowding and inefficiency. The marginal product of labor declines, meaning more input is needed to produce each additional unit. Consequently, marginal cost rises. From the figure 16.4 it is clear that the marginal cost curve first slopes downward, reaches a minimum point, and then slopes upward.

Average Fixed Cost (AFC)

Average Fixed Cost (AFC) refers to the fixed cost per unit of output produced. It shows how much of the firm's fixed expenses are allocated to each unit of output. Since fixed costs remain constant in the short run, AFC depends entirely on the level of output. The formula for Average Fixed Cost is:

$$AFC = F/q$$

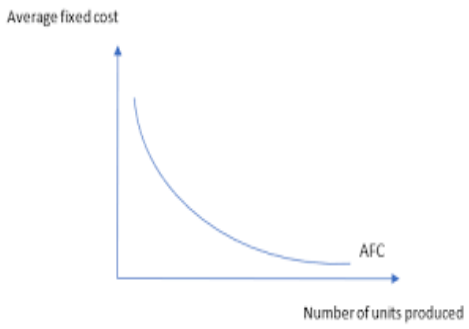
Where,

F = Fixed Cost

q = Quantity of output

Because fixed cost does not change with output, AFC decreases continuously as output increases. This happens because the same total fixed cost is spread over a larger number of units. This process is known as the spreading effect of fixed costs. AFC can be graphically represent as shown in figure 16.5.

Figure 16.5: AFC Curve



From the figure 16.5 we can see that the AFC curve slopes downward continuously from left to right. It declines rapidly at low levels of output. It declines gradually at higher levels of output. It approaches the horizontal axis (X-axis) but never touches it. The reason it never touches the axis is that fixed cost can never become zero in the short run. No matter how large the output becomes, there will always be some positive fixed cost, even if it becomes very small per unit.

Average Variable Cost (AVC)

Average Variable Cost (AVC) refers to the variable cost per unit of output. It shows how much variable expense is incurred, on average, to produce each unit of output in the short run. Since variable costs change with output, AVC reflects the efficiency of the firm's use of variable inputs such as labor, raw materials, and electricity. The formula for Average Variable Cost is:

$$AVC = VC/q$$

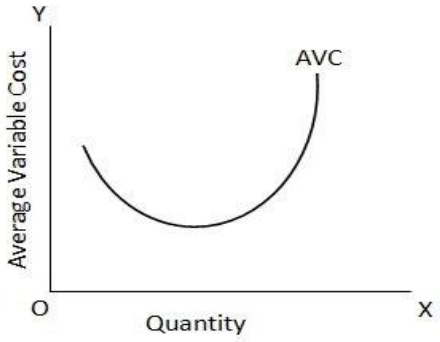
Where,

VC = Variable Cost

q = Quantity of output

Average Variable Cost measures the per-unit variable expense of production. It reflects the efficiency of variable input usage and it helps to determine the short-run supply curve of a firm. Graphically AVC curve is shown in figure 16.6.

Figure 16.6: AVC Curve



The AVC curve is typically U-shaped, and this shape is explained by the law of diminishing marginal returns.

Initial Stage (Decreasing AVC): At low levels of output, AVC decreases. This happens because fixed inputs such as machinery and plant are underutilized at first. As more variable inputs (like labor) are employed, better coordination, specialization, and efficient use of fixed factors occur. As a result, the cost per unit falls.

Later Stage (Increasing AVC): After a certain level of output, diminishing returns begin. Since at least one factor (like capital) is fixed in the short run, adding more variable inputs leads to overcrowding and inefficiency. The marginal product of labor declines, meaning more variable input is required to produce additional units. Consequently, AVC rises.

Thus, the AVC curve first slopes downward, reaches a minimum point, and then slopes upward, forming a U-shape. It is clear from the figure 16.6.

Average Total Cost (AC)

Average Total Cost (ATC), also known as Average Cost (AC), refers to the total cost per unit of output. It shows how much it costs, on average, to produce one unit of a good. This concept combines both fixed and variable costs into a single per-unit measure. The formula for Average Total Cost is:

$$AC = C/q$$

Where,

C = Total Cost

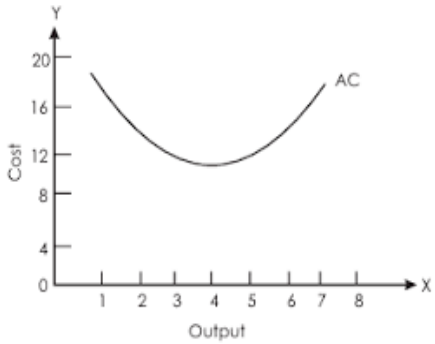
q = Quantity of output

Since total cost is the sum of fixed and variable costs, AC can also be expressed as:

$$AC = AFC + AVC$$

This equation shows that average total cost is made up of two components Average Fixed Cost (AFC) and Average Variable Cost (AVC). Graphical representation of AC is shown in the figure 16.7.

Figure 16.7: AC Curve



The AC curve is typically U-shaped, and this shape is due to the combined behaviour of AFC and AVC.

Initial Stage that is Falling AC: At low levels of output, AC decreases. This happens for two main reasons firstly, AFC falls continuously as output increases because fixed cost is spread over more units. Secondly, AVC initially falls due to better efficiency, specialization, and improved use of fixed inputs. The combined effect of falling AFC and falling AVC causes AC to decline.

Minimum Point: AC reaches its lowest point when the reduction in AFC is just offset by the rise in AVC. This is the most efficient scale of production in the short run.

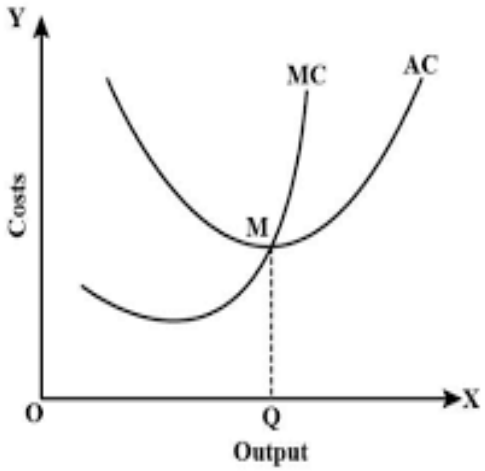
Later Stage (Rising AC): After the minimum point, AC increases. Although AFC continues to decline, AVC rises sharply due to diminishing marginal returns. The increase in AVC outweighs the fall in AFC, causing AC to rise.

Relationship Between Marginal Cost and Average Cost

The relationship between Marginal Cost (MC) and Average Cost (AC) follows an important mathematical principle that explains the behaviour of cost curves in the short run. When marginal cost is less than average cost (MC < AC), the average cost falls because the additional unit produced costs less than the existing average, thereby pulling the average downward.

When marginal cost is greater than average cost (MC > AC), the average cost rises because the additional unit costs more than the current average, pushing the average upward. When marginal cost equals average cost (MC = AC), the average cost is at its minimum point, since at that stage the marginal unit neither increases nor decreases the average. This relationship is similar to the connection between marginal product and average product in production theory, where the marginal value influences the movement of the average value. Graphically, this principle is represented by the marginal cost curve intersecting the average cost curve exactly at its lowest point, which reflects the most efficient scale of production in the short run as shown in the figure 16.8.

Figure 16.8: Marginal Cost and Average Cost



Importance of Short-Run Cost Analysis

Short-run cost analysis plays a vital role in managerial decision-making and economic planning. In the short run, at least one factor of production is fixed, which makes cost behaviour particularly important for understanding how firms respond to changes in output and market conditions. By studying different cost concepts such as fixed cost, variable cost, average cost, and marginal cost, firms can make rational and profit-oriented decisions.

1. Short-run cost analysis helps firms determine the optimal level of output. A firm maximizes profit where marginal cost equals marginal revenue. By analysing cost curves, especially the marginal cost curve, firms can identify the most efficient quantity to produce in order to maximize profits or minimize losses.

2. It assists in pricing decisions. In competitive markets, firms are price takers, and they compare market price with average and marginal costs to decide how much to produce. In imperfect markets, cost information helps firms set prices that cover costs and generate reasonable profit margins. Understanding average total cost ensures that prices are not set below sustainable levels in the long run.
3. Short-run cost analysis helps firms decide whether to continue production or shut down temporarily. The minimum point of the Average Variable Cost curve represents the shutdown point. If the market price falls below AVC, the firm cannot cover its variable costs and should shut down to minimize losses. If price is above AVC but below Average Total Cost (AC), the firm may continue operating in the short run while incurring a temporary loss.
4. Cost analysis enables firms to estimate profitability. By comparing total revenue with total cost, or price with average cost, firms can determine whether they are earning supernormal profit, normal profit, or incurring losses. This evaluation is essential for financial planning and business sustainability.
5. It helps firms understand production efficiency. The shape of cost curves reflects the impact of economies of specialization and diminishing returns. A falling average cost indicates improved efficiency, while rising costs signal inefficiencies and overutilization of fixed inputs.

Short-run cost analysis is a fundamental concept in microeconomics. It explains how production costs behave when at least one factor of production is fixed. By understanding fixed cost, variable cost, total cost, marginal cost, and average costs, firms can make rational profit-maximizing decisions. The relationships among cost curves—particularly between marginal and average costs—provide powerful tools for analysing firm behaviour. The U-shaped nature of cost curves reflects the underlying law of diminishing returns. In conclusion, short-run cost theory equips managers and economists with the analytical framework necessary to understand production efficiency, pricing strategy, shutdown decisions, and profit maximization. A thorough understanding of these concepts ensures better managerial decisions and efficient resource allocation in competitive markets.

Multiple Choice Questions

1. What characterizes fixed costs in the short run?

A. They change directly with the level of output.

B. They remain constant regardless of the production level.

C. They can be adjusted by changing the labor force.

D. They increase as production increases.

Answer: B. They remain constant regardless of the production level.

2. Which of the following statements about variable costs is true?

A. Variable costs do not change with output.

B. Variable costs decrease when output decreases.

C. Variable costs are fixed in the short run.

D. Variable costs remain constant regardless of output levels.

Answer: B. Variable costs decrease when output decreases.

3. What is the relationship between marginal cost and average cost at the minimum point of the average cost curve?

A. MC is greater than AC.

B. MC is less than AC.

C. MC equals AC.

D. MC is unrelated to AC.

Answer: C. MC equals AC.

4. Why is short-run cost analysis important for firms?

A. It helps determine which factors of production are fixed.

B. It assists in identifying the optimal level of output for maximizing profit.

C. It guarantees that all variable costs will be covered.

D. It eliminates the need for financial planning.

Answer: B. It assists in identifying the optimal level of output for maximizing profit.

5. What happens if the market price falls below the minimum point of the Average Variable Cost (AVC) curve?

A. The firm should continue operating to maximize losses.

B. The firm should shut down temporarily to minimize losses.

C. The firm will earn supernormal profits.

D. The firm can afford to expand production.

Answer: B. The firm should shut down temporarily to minimize losses.

III. Production Functions and the Shape of Cost Curves

In microeconomic theory, the study of production and cost forms the foundation of firm behaviour. Every firm aims to maximize profit, and to do so, it must clearly understand how production decisions translate into costs. The relationship between inputs and outputs is described by the production function, and this function ultimately determines the shape of cost curves in the short run. In the short run, at least one factor of production remains fixed while other factors are variable. Because of this constraint, firms experience important economic phenomena such as diminishing marginal returns, which directly influence the shape of variable cost, marginal cost, and average cost curves. Additionally, government policies such as taxation significantly affect firms' cost structures. Taxes may alter marginal and average costs, influencing production decisions, pricing, and profitability.

A production function shows the maximum quantity of output that can be produced using given quantities of inputs, assuming efficient use of resources. It reflects technological knowledge and input combinations available to the firm. In symbolic form:

$$q = f(L, K)$$

Where,

q = Output

L = Labor

K = Capital

In the short run, capital (K) is fixed. Therefore, output depends only on labor. Therefore

$$q = f(L, \bar{K}).$$

The Total Product (TP) of labor shows the total output produced at different levels of labor, holding capital fixed. Initially, as labor increases output increases at an increasing rate and workers specialize and use fixed capital more efficiently. Eventually Output increases at a decreasing rate, workers crowd the fixed capital and the Law of Diminishing Marginal Returns operates. The Law of Diminishing Marginal Returns states that when additional units of a variable input are added to a fixed input, the marginal product of the variable input eventually declines. This law is central to understanding cost curves in the short run.

The production function determines the shape of a firm's cost curves. The production function shows the amount of inputs needed to produce a given level of output. The firm calculates its cost by multiplying the quantity of each input by its price and summing the costs of the inputs. Costs arise from the use of inputs. If a firm produces output using capital and labor, and its capital is fixed in the short run, the firm's variable cost is its cost of labor. Its labor cost is the wage per hour, w, times the number of hours of labor, L, employed by the firm:

$$VC = wL$$

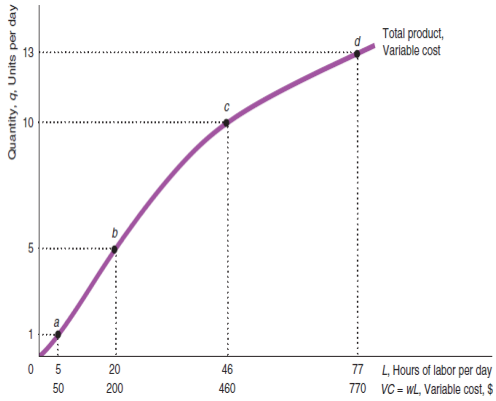
In the short run, when the firm's capital is fixed, the only way the firm can increase its output is to use more labor. If the firm increases its labor enough, it reaches the point of diminishing

marginal return to labor, at which each extra worker increases output by a smaller amount. We can use this information about the relationship between labor and output—the production function—to determine the shape of the variable cost curve and its related curves.

Shape of the Variable Cost Curve

If input prices are constant, the production function determines the shape of the variable cost curve. We illustrate this relationship for the firm in the figure 16. 9.

Figure 16. 9: Variable Cost Curve



The firm faces a constant input price for labor, the wage, of $10 per hour. The total product of labor curve in figure 16.9 shows the firm's short-run production function relationship between output and labor when capital is held fixed. At point a, the firm uses 5 hours of labor to produce 1 unit of output. At point b, it takes 20 hours of labor to produce 5 units of output. Here, output increases more than in proportion to labor output rises 5 times when labor increases 4 times. In contrast, as the firm moves from b to *c*, output increases less than in proportion. Output doubles to 10 as result of increasing labor from 20 to 46—an increase of 2.3 times. The movement from c to d results in even a smaller increase in output relative to labor. This flattening of the total product curve at higher levels of labor reflects diminishing marginal returns to labor. This curve shows both the production relation of output to labor and the variable cost relation of output to cost. Because each hour of work costs the firm $10, we can use an alternative measure for the horizontal axis in the figure 16.9 the firm's variable cost, which is its cost of labor. To produce 5 units of output takes 20 hours of labor, so the firm's variable cost is $200. By using the variable cost labels on the horizontal axis, the total product of labor curve becomes the variable cost curve. As output increases, the variable cost increases more than proportionally due to the

diminishing marginal returns. Because the production function determines the shape of the variable cost curve, it also determines the shape of the marginal, average variable, and average cost curves.

Shape of the Marginal Cost Curves

The marginal cost is the change in variable cost as output increases by one unit:

$$MC = \Delta VC/\Delta q$$

In the short run, capital is fixed, so the only way the firm can produce more output is to use extra labor. The extra labor required to produce one more unit of output is

$$\Delta L/\Delta q$$

The extra labor costs the firm w per unit, so the firm's cost rises by $w(\Delta L/\Delta q)$. As a result, the firm's marginal cost is

$$MC = \Delta VC/\Delta q = w\ (\Delta L/\Delta q)$$

The marginal cost equals the wage times the extra labor necessary to produce one more unit of output. To increase output by one unit from 5 to 6 units takes 4 extra workers in the figure 16.9. If the wage is $5 per hour, the marginal cost is $20. The information about how much extra labor we need to produce one more unit of output comes from the production function. The marginal product of labor is

$$MPL = \Delta q/\Delta L$$

Thus, the extra labor we need to produce one more unit of output, $\Delta L/\Delta q$, is 1/MPL, so the firm's marginal cost is

$$MC = w/\ MPL$$

If the firm is producing 5 units of output, it takes 4 extra hours of labor to produce 1 more unit of output in the figure 16.9.

At low levels of labor, the marginal product of labor commonly rises with additional labor because extra workers help the original workers and they can collectively make better use of

the firm's equipment. As the marginal product of labor rises, the marginal cost falls. Finally, as the number of workers increases, workers must share the fixed amount of equipment and may get in each other's way, so the marginal cost curve slopes upward because of diminishing marginal returns to labor. Thus, the marginal cost first falls and then rises.

Shape of Average Cost Curves

In the short run, the shape of the average cost curves is fundamentally determined by the production process, particularly by the law of diminishing marginal returns to labor. Since capital is fixed in the short run and labor is the only variable input, the behaviour of labor productivity directly influences the behaviour of cost curves. The average variable cost (AVC) is defined as

$$AVC = VC/q$$

When labor is the only variable input and the wage rate (w) is constant, variable cost is equal to wL, where L represents units of labor. Substituting this into the formula gives

$$AVC = wL/q$$

Because the average product of labor (APL) is defined as q/L, the expression can be rearranged to show that

$$AVC = w/APL$$

This relationship clearly demonstrates that average variable cost is equal to the wage rate divided by the average product of labor. Therefore, with a constant wage rate, AVC moves in the opposite direction of APL.

In general, during the early stages of production, as more labor is employed, workers can specialize and make better use of fixed capital, which causes the average product of labor to rise. As APL rises, AVC falls. But, beyond a certain level of employment, the fixed capital becomes crowded and workers begin to interfere with each other's tasks. This leads to diminishing marginal returns, causing the average product of labor to decline. As APL declines, AVC rises. Because APL first increases and then decreases, AVC correspondingly first falls

and then rises, giving it a U-shaped curve. The average fixed cost (AFC), behaves differently. It is defined as

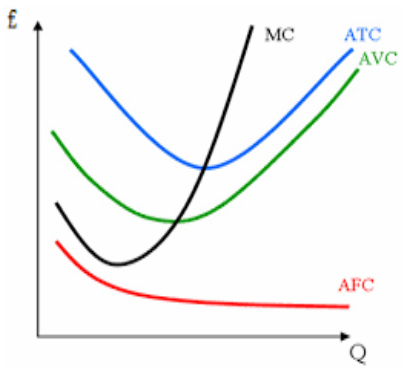
$$AFC = FC/q$$

where fixed cost remains constant regardless of output. As output increases, the same fixed cost is spread over a larger number of units, causing AFC to continuously decline. The AFC curve therefore slopes downward throughout and approaches zero as output increases, although it never actually reaches zero. When we combine average fixed cost and average variable cost, we obtain average cost (AC), which is defined as

$$AC = AFC + AVC$$

Since AVC is U-shaped and AFC is continuously falling, the average cost curve reflects the combined influence of these two components. At low levels of output, average fixed cost is very high because output is small, and average variable cost is falling due to rising productivity. As a result, average cost falls sharply. In this stage, both falling AFC and falling AVC contribute to the decline in AC, making the AC curve fall more steeply than the AVC curve. As output increases further, AVC eventually reaches its minimum point and begins to rise due to diminishing marginal returns, while AFC continues to fall but at a slower rate. At higher levels of output, AFC becomes very small because fixed cost is spread over many units. Consequently, the gap between AC and AVC becomes narrower. Since AFC approaches zero as output expands, the AC curve and AVC curve move closer together at high levels of output. Eventually, the rising AVC outweighs the small decline in AFC, causing AC to rise. Therefore, the average cost curve is also U-shaped as shown in the figure 16.10.

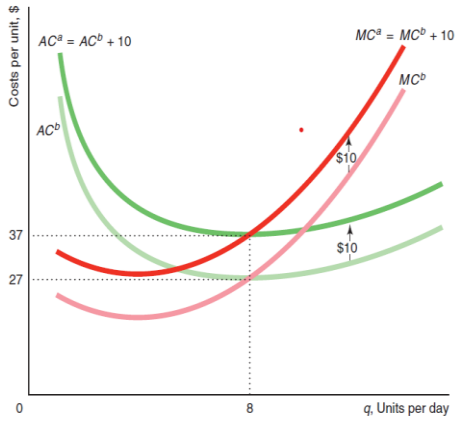
Figure 16. 10: Variable Cost Curve



IV. Effects of Taxes on Costs

Taxes imposed on firms play an important role in shaping their cost structure because they directly influence production expenses. When the government levies a tax, it can shift some or all of the firm's marginal and average cost curves depending on the nature of the tax. Consider first a specific tax**,** which is a fixed amount charged per unit of output.

Figure 16. 11: Effect of Specific Tax



Suppose the government imposes a specific tax of $10 per unit produced. Since this tax varies directly with output, it becomes part of the firm's variable cost rather than its fixed cost. As a result, it increases variable cost, average variable cost (AVC), average cost (AC), and marginal cost (MC), but it does not affect average fixed cost (AFC). At every level of output, the firm must now pay an additional $10 for each unit it produces. This means that the average variable cost increases by exactly the amount of the tax. For instance, if the firm's average variable cost before the tax (AVCb) is $21 at an output level of 8 units, the imposition of a $10 per-unit tax raises the after-tax average variable cost (AVCa) to $31.

In general terms, the after-tax average variable cost equals the before-tax average variable cost plus the tax per unit; that is, AVCa = AVCb + 10. Because average cost AC is the sum of average variable cost and average fixed cost, and since the tax does not affect fixed cost, the entire increase in AVC translates directly into an equal increase in AC. For example, if the firm's average cost before tax ACb is $27, the new average cost after tax ACa becomes $37. The specific tax also raises marginal cost because each additional unit produced now requires the firm to pay not only the production cost of that unit but also the additional $10 tax. If the firm's marginal cost before tax (MCb) is $27 at 8 units of output, the marginal cost after tax

(MCa) becomes $37. Thus, MCa = MCb + 10. Graphically, this means that the marginal cost curve, the average variable cost curve, and the average cost curve all shift upward by exactly $10 at every quantity of output as shown in the figure 16.11. But, the average fixed cost curve remains unchanged because the tax does not alter fixed costs. An important implication of this uniform upward shift is that the output level at which average cost is minimized does not change. Since both the marginal cost and average cost curves shift upward by the same vertical amount, their point of intersection—where marginal cost equals average cost and average cost is at its minimum—remains at the same quantity of output. Therefore, although a specific tax increases the firm's cost of production at every output level, it does not alter the output at which average cost is minimized.

In contrast, a franchise tax, also known as a lump-sum tax or business license fee, affects costs differently. A franchise tax is a fixed payment that a firm must make for the right to operate, regardless of how much output it produces. Because this tax does not vary with output, it increases only the firm's fixed cost and has no effect on variable cost. As a result, average fixed cost (AFC) increases, and since average cost (AC) is the sum of AFC and AVC, the AC curve also shifts upward. But, because variable cost is unaffected, neither the average variable cost curve nor the marginal cost curve changes.

This distinction is crucial for understanding firm behaviour. A specific tax increases marginal cost and therefore influences production decisions, since firms typically determine output where marginal revenue equals marginal cost. A franchise tax, on the other hand, does not change marginal cost and therefore does not directly affect the firm's output decision in the short run; it simply reduces profit by increasing fixed cost. In summary, while both types of taxes raise a firm's overall cost burden, a specific tax alters variable and marginal costs and shifts several cost curves upward, whereas a franchise tax affects only fixed costs and shifts only the average cost curve without changing marginal cost.

The production function forms the backbone of cost analysis in the short run. Because capital is fixed, output expansion depends entirely on labor. As labor increases, the Law of Diminishing Marginal Returns eventually operates, causing marginal product to fall. This decline in marginal product leads directly to rising marginal cost. Thus: Variable cost rises at an increasing rate, Marginal cost is U-shaped, Average variable cost is U-shaped. Average cost is U-shaped. Average fixed cost continuously declines. But the Government taxation alters cost structures: A specific tax shifts MC, AVC, and AC upward. A franchise tax shifts only AC and

AFC upward. The cost-minimizing output remains unchanged under a specific tax. Understanding these relationships is essential for managerial decision-making, policy analysis, and economic theory. The theory of short-run costs provides the bridge between production technology and firm behaviour in markets.

Multiple Choice Questions

1.What does the production function describe in microeconomic theory?

A. The maximum quantity of output that can be produced using given quantities of inputs

B. The total cost incurred by a firm at different levels of output

C. The relationship between fixed and variable costs in the long run

D. The effect of government policies on consumer behaviour.

Answer: A. The maximum quantity of output that can be produced using given quantities of inputs.

2. In the short run, which factor of production remains fixed according to the production function?

A. Labor

B. Capital

C. Output

D. Variable Costs.

Answer: B. Capital.

3. The Law of Diminishing Marginal Returns states that:

A. Increasing the number of workers will always lead to higher output indefinitely

B. Adding more units of a variable input to a fixed input eventually leads to a decline in the marginal product

C. Marginal costs decrease as production increases

D. Total costs will always rise with increased production.

Answer: B. Adding more units of a variable input to a fixed input eventually leads to a decline in the marginal product.

4. How does a specific tax affect a firm's marginal cost in the short run?

A. It decreases marginal cost at every level of output

B. It has no effect on marginal cost

C. It increases marginal cost at every level of output

D. It only affects fixed costs without influencing marginal cost.

Answer: C. It increases marginal cost at every level of output.

5. Which statement best reflects the effect of a lump-sum tax on a firm's production decisions?

A. It increases marginal costs and leads to a reduction in output

B. It alters the variable costs and shifts the marginal cost curve

C. It affects only fixed costs and does not change production levels

D. It decreases overall profit without affecting fixed costs.

Answer: C. It affects only fixed costs and does not change production levels.